

Exercise 1

Problem: We aim to evaluate the throughput and the time delay characterizing the TDMA and FDMA multiple-access techniques, for the following network parameters:

- Number of stations, $N = 16$
- Offered traffic per station, $\lambda = 40 \text{ Ul/s}$
- IU packet length, $L = 1000 \text{ byte}$
- Network capacity, $C = 10 \text{ Mbps}$

Solution: The network throughput is the same regardless of the access protocol, assuming that no IU is lost. It is equal to:

$$\rho = \lambda \frac{NL}{C} = 40 \frac{16 \cdot 8 \cdot 1000}{10 \cdot 10^6} = 0.512$$

Starting from the network throughput, we can calculate the average delay in TDMA and FDMA:

$$D_{TDMA} = \frac{L}{C} + \frac{\rho}{2(1-\rho)} \frac{NL}{C} + \frac{1}{2} \frac{NL}{C} = (0.8 + 6.71 + 6.4) \cdot 10^{-3} = 13.91ms$$

$$D_{FDMA} = \frac{\rho}{2(1-\rho)} \frac{NL}{C} + \frac{NL}{C} = (6.71 + 12.8) \cdot 10^{-3} = 19.51ms$$

Exercise 2

Problem: A TDMA multiple access system uses $N = 10$ time slot, a guard time $T_g = 200[\mu s]$, data packets composed of $D = 180 \text{ [bit]}$ and $H = 20 \text{ [bit]}$ of header, and a frame time $T_T = 10[ms]$. Calculate the transmission rate W and the net data rate V of each channel.

Solution: The size of a slot is $k = D + H = 180 + 20 = 200 \text{ [bit]}$. The slot time is:

$$T_S = \frac{T_T}{N} = \frac{10 \text{ ms}}{10} = 1ms$$

The transmission time of the data part of the slot is:

$$T_D = T_S - T_g = 1ms - 200\mu s = 0.8ms$$

The multiplexed data rate is:

$$W = \frac{k}{T_D} = \frac{200bit}{0.8ms} = 250kbps$$

The net channel rate is:

$$V = \frac{D}{T_T} = \frac{180bit}{10ms} = 18kbps$$

Exercise 3

Problem: A centralized time-division multiple access (TDMA) system is characterized by a frame with slot duration $T_S = 10\mu s$ and a minimum guard time $T_G = 2\mu s$. The system serves $N = 8$ users and has a multiplexed signal transmission rate of $C = 1Mbps$. It is required to:

1. Determine the number of bits of each tributary stream transmitted in each slot, n
2. Determine the maximum possible input rate for each tributary stream, r

Solution:

1. $n = (10 - 2)\mu s \cdot 1Mbps = 8bit$
2. $r = \frac{8bit}{8 \cdot 10\mu s} = 100kbps$

Exercise (ALOHA Protocol)

Problem: Consider a network based on ALOHA. The packet duration is T . Assume that the traffic on the channel (average number of transmissions in time T) is $1.649 = \sqrt{e}$ times the number of successfully transmitted frames. Compute the network throughput.

Solution: The throughput formula S as a function of the channel traffic G for ALOHA is $S = Ge^{-2G} \iff G/S = e^{2G}$. From the problem data:

$$G/S = 1.649$$

$$e^{2G} = 1.649 \Rightarrow 2G = \ln(1.649) \Rightarrow 2G = \ln(\sqrt{e}), G = \frac{1}{4}$$

It follows that the throughput S is:

$$S = Ge^{-2G} = \frac{1}{4 \cdot 1.649} = 0.151$$

Exercise (Collision Evaluation)

Problem: There are 3 stations communicating over a shared medium using the ALOHA protocol. Assuming zero propagation time, transmission rate $R = 2$ Mbps, and IU length $L = 1000$ byte, the following transmissions occur:

- Station A: $t_A = 400ms$
- Station B: $t_B = 410ms$
- Station C: $t_C = 405ms$ Do collisions occur?

Solution: $T_{UI} = \frac{L}{R} = \frac{1000 \cdot 8}{2 \cdot 10^6} = 4ms$. No, collisions do not occur.

Exercise (Collision Resolution)

Problem: Considering the previous setup, now with a transmission rate $R = 1$ Mbps. Do collisions occur? For how long do the UIs overlap? Assume stations retransmit after collisions based on $\epsilon = (seed) \cdot 2N_c T_{UI}$, where the seed sequence is $[2, 1, 1, 3, 2]$ and N_c is the number of consecutive collisions. Determine the success instants.

Solution: $T_{UI} = \frac{L}{R} = 8ms$.

- **Station A:** First collision, $\epsilon = 2 \cdot 2 \cdot 1 \cdot 8 = 32ms$. Retransmission instant: $408 + 32 = 440$ ms.
- **Station B:** First collision, $\epsilon = 1 \cdot 2 \cdot 1 \cdot 8 = 16ms$. Retransmission instant: $418 + 16 = 434$ ms.
- **Station C:** First collision, $\epsilon = 1 \cdot 2 \cdot 1 \cdot 8 = 16ms$. Retransmission instant: $413 + 16 = 429$ ms. Second collision, $\epsilon = 3 \cdot 2 \cdot 2 \cdot 8 = 96ms$. Retransmission instant: $413 + 48 = 509$ ms.

Exercise: Throughput of Slotted ALOHA with a Finite Number of Stations

Problem: Calculate the value of the maximum throughput of the slotted ALOHA protocol as a function of the number of stations N in the network.

Solution: Similarly to the case of infinite stations, we compute the first derivative of the throughput $S = G(1 - \frac{G}{N})^{N-1}$ and set it equal to 0:

$$S' = \frac{d(G(1 - \frac{G}{N})^{N-1})}{dG} = (1 - \frac{G}{N})^{N-1} + \left[-\frac{1}{N}G(N-1)(1 - \frac{G}{N})^{N-2} \right] = 0$$

$$(1 - \frac{G}{N})^{N-1} - \frac{G(N-1)}{N}(1 - \frac{G}{N})^{N-2} = 0$$

Setting $(1 - \frac{G}{N}) = X$, we get:

$$X^{N-1} \left[1 - \frac{G(N-1)}{N} X^{-1} \right] = 0$$

$$1 - \frac{G(N-1)}{N} \cdot \frac{1}{(1 - \frac{G}{N})} = 0$$

$$\frac{G(N-1)}{N-G} = 1 \implies G(N-1) = N-G \implies GN - G = N - G \implies GN = N \implies G$$

The maximum throughput for $G = 1$ is:

$$S_{max}(G = 1) = \left(1 - \frac{1}{N}\right)^{N-1}$$

Exercise: Slotted ALOHA Collision Resolution

Problem: There are 3 stations communicating using the slotted ALOHA protocol. Assume that the propagation time is zero and that the UI length is fixed and occupies the channel for the entire duration of a time slot. Each station is assigned a seed used to generate random numbers as follows:

Station A	Station B	Station C
27	13	9

All stations start transmitting in the first slot. After a collision, stations retransmit after X slots, where $X = \lfloor \sqrt{\text{seed}_{A,B,C} \cdot N_c} \rfloor$ and N_c is the number of consecutive collisions. Determine in which slot the 3 stations are able to successfully transmit.

Solution:

- **Station A:** * First collision: $X = \lfloor \sqrt{27 \cdot 1} \rfloor = 5$
- **Station B:**
 - First collision: $X = \lfloor \sqrt{13 \cdot 1} \rfloor = 3$
 - Second collision: $X = \lfloor \sqrt{13 \cdot 2} \rfloor = 5$
- **Station C:**
 - First collision: $X = \lfloor \sqrt{9 \cdot 1} \rfloor = 3$
 - Second collision: $X = \lfloor \sqrt{9 \cdot 2} \rfloor = 4$

Exercise 1

A smart building has 50 light bulbs and 5 switches. Each switch is configured to publish to a unique "Zone" group address. 10 bulbs are subscribed to each Zone. If every switch sends an "ON" command (1 message/switch) simultaneously, and each relay node in the mesh has a retransmission count of 3, calculate the total number of message transmissions occurring in the network.

Solution

For each switch, the number of original messages is:

$$5 \text{ switch} \times 1 \text{ message} = 5 \text{ original messages}$$

Each message is sent once and then retransmitted 3 times:

$$\text{Total transmissions} = 5 \times (3 + 1) = 20 \text{ (1 original + 3 retransmissions)}$$

Exercise 2

A Friend Node has a buffer of 1000 bytes for cached messages. It supports 5 LPNs. If each LPN receives an average of 4 messages per hour, and each message is 20 bytes, how long can the Friend Node cache messages for all LPNs before the buffer overflows?

Solution

Total message size for one round of messages:

Since every LPN receives 4 messages/h, we have:

$$5 \text{ LPN} \times 4 \text{ messages/h} = 20 \text{ messages/h}$$

Due to the fact that every message is 20 byte, the Friend Node collects:

$$20 \frac{\text{messages}}{\text{h}} \times 20 \text{ byte} = 400 \text{ bytes/h}$$

How long does the buffer last?

$$\frac{1000}{400} = 2.5 \text{ hours}$$

Exercise 3

The standard Bluetooth Classic codec (SBC) often runs at 345 kbps for high-quality stereo. The new LC3 codec can achieve equivalent or better perceived quality at roughly half that bitrate (e.g., 160 kbps). If a smartphone streams audio for 2 hours:

1. Calculate the total data transmitted (in Megabytes) using SBC at 345 kbps.

2. Calculate the total data transmitted using LC3 at 160 kbps.
3. What is the percentage of data reduction when switching to LE Audio?

Solution

Data transmitted with:

SBC:

$$345 \text{ kbps} \times 7200 \text{ s} = 2.484.000 \text{ kbits} \approx 310.5 \text{ MB}$$

LC3:

$$160 \text{ kbps} \times 7200 \text{ s} = 1.152.000 \text{ kbits} \approx 144 \text{ MB}$$

The percentage of data reduction is given as:

$$310.5 - 144 = 166.5 \text{ MB} \quad \frac{166.5}{310.5} \times 100 \approx 53.6\%$$

Exercise 4

In a ZigBee mesh network for an industrial automation system, the following specifications are provided:

- Total number of nodes: 45 (1 coordinator + 12 FFD routers + 32 RFD end devices)
- Data rate: 250 kbps
- Data packet size: 80 bytes
- ACK packet size: 15 bytes
- Propagation time per hop: 0.5 ms
- Processing time per node: 2 ms
- Maximum allowed hops: 8
- Network size: 200 m x 150 m
- Average distance between adjacent nodes: 25 m

1. Calculate the total latency for a packet crossing 5 hops.
2. Calculate the total transmission time for a packet with ACK (SIF = 192 μ s).

Solution 1 Number of nodes processing = 5 hops = 5 intermediate routers + 1 destination = 6 nodes
 Total propagation time = 5 hops $\times$ 0.5 ms/hop = 2.5 ms
 Total processing time = 6 nodes $\times$ 2 ms/node = 12 ms
 Total latency = 2.5 ms + 12 ms = 14.5 ms

Solution 2

Packet duration:

$$80 \text{ bytes} \times 8 \text{ bits/byte} = 640 \text{ bits} \quad \text{Time} = \frac{640 \text{ bits}}{250,000 \text{ bits/s}} = 0.00256 \text{ s} = 2.56 \text{ ms}$$

ACK packet duration:

$$15 \text{ bytes} \times 8 \text{ bits/byte} = 120 \text{ bits} \quad \text{Time} = \frac{120 \text{ bits}}{250,000 \text{ bits/s}} = 0.00048 \text{ s} = 0.48 \text{ ms}$$

$$\text{SIF} = 192 \mu\text{s} = 0.192 \text{ ms} \quad \text{Total time} = 2.56 \text{ ms} + 0.192 \text{ ms} + 0.48 \text{ ms} = 3.232 \text{ ms}$$

Exercise 5

Calculate the maximum network throughput.

- How many packets can be transmitted per second on a single channel?
- If 10 nodes transmit simultaneously, what is the effective throughput per node?

Solution Packets per second on a channel: $1/0.003232 \text{ s} = 309 \text{ packets/second}$

Theoretical throughput: $309 \text{ packets/s} \times 640 \text{ bits/packet} = 197,760 \text{ bit/s} \approx 198 \text{ kbps}$
Efficiency = $198 \text{ kbps}/250 \text{ kbps} = 79.2\%$

With 10 nodes transmitting simultaneously: Throughput per node = $198 \text{ kbps}/10 = 19.8 \text{ kbps/node}$

Exercise 6

Calculate energy consumption per transmission (Transmission power: 100 mW, Transmission duration: 3.232 ms).

- If a node transmits 50 times a day, how many Joules/day does it consume?

Solution Power = 100 mW = 0.1 W Transmission time = 3.232 ms = 0.003232 s

Energy per transmission = $0.1 \text{ W} \times 0.003232 \text{ s} = 0.0003232 \text{ J} = 0.3232 \text{ mJ}$
Daily energy (50 transmissions) = $0.3232 \text{ mJ} \times 50 = 16.16 \text{ mJ/day} = 0.01616 \text{ J/day}$

Exercise 7

Verify if the network is congested:

- 32 sensors send data every 30 seconds.
- Transmission duration per packet: 3.232 ms.
- How much time is the channel occupied in one minute?
- Is the channel saturated ($>100\%$ utilization)?

Solution 32 sensors send every 30 seconds. Transmissions per minute =
 $32 \text{ sensors} \times \left(\frac{60 \text{ s}}{30 \text{ s}}\right) = 64 \text{ transmissions/minute}$ Time occupied per minute:
 $64 \text{ transmissions} \times 3.232 \text{ ms} = 206.848 \text{ ms} = 0.206848 \text{ s}$ Channel utilization =
 $\frac{0.206848 \text{ s}}{60 \text{ s}} = 0.00345 = 0.345\%$ Channel is NOT congested (only 0.345% utilization)

Exercise: LoRa Network Latency Analysis

Problem: Let us consider a LoRa network with one device and a gateway transmitting frames of length $L = 13$ byte. Initially, the gateway sends a frame and then, after 56 ms, the device sends another frame. Assuming a data rate of $R = 1\text{Kbps}$, a distance from the device to the gateway of $d = 10$ km and no other delay except those of transmission and propagation, we want to know how much it takes to transmit and receive both frames, assuming devices of class A and C?

Solution: In class C, the gateway and the device transmit the frames as soon as possible. Therefore, the total delay will be:

$$D_C = 2 \times (D_t + D_p) \approx 2D_t = 208 \text{ ms}$$

Where the transmission delay D_t is:

$$D_t = \frac{8 \times L}{R} = \frac{8 \times 13}{1000} = 104 \text{ ms}$$

And the propagation delay D_p is:

$$D_p = \frac{d}{c} = \frac{10 \times 10^3}{3 \times 10^8} = 33.34 \mu\text{s}$$

In class A, the gateway transmits only in response to the frame sent by the device. This corresponds to a total delay equal to:

$$D_A = D_C + 56 \text{ ms} = 264 \text{ ms}$$

Compared to Class C, Class A provides lower power consumption but results in increased latency.

Exercise: Pub/Sub Transmission Delay

Problem: Let us consider the following scenario where a sensor event is transmitted to a gateway, then a broker, and finally to an application. Assuming a lossless network in which the only delay is transmission delay, we want to know how long it takes to transmit a 200-byte packet from the sensor to the application with $R_{\text{sensor}} = 100 \text{ Kbps}$ and $R_{\text{GW}} = R_{\text{broker}} = 1 \text{ Gbps}$. Moreover, what happens if

the transmission rate between the gateway, the broker, and the application is also 100 Kbps?

Solution: Neglecting propagation, processing, and queuing delays, the packet is transmitted by the sensor and retransmitted by the gateway and the broker. For the first case, $L = 200$ byte:

$$D = \left(\frac{L}{R_{Sensor}} + \frac{L}{R_{GW}} + \frac{L}{R_{broker}} \right) = 200 \times 8 \left(\frac{10^{-3}}{100} + \frac{2 \times 10^{-9}}{1} \right) = 16 \text{ ms}$$

For the second case, where $R_{sensor} = R_{GW} = R_{broker} = 100$ Kbps:

$$D = \left(\frac{L}{R_{Sensor}} + \frac{L}{R_{GW}} + \frac{L}{R_{broker}} \right) = 200 \times 8 \times \frac{3 \times 10^{-3}}{100} = 48 \text{ ms}$$

Exercise: REST Network Delay

Problem: Let us consider a lossless REST network. Assuming that the only delay is transmission delay, calculate the time needed for the application to receive a 200-byte response packet, considering a 100-byte request packet,

$R_{sensor} = 100$ Kbps, $R_{GW} = R_{broker} = 1$ Gbps, and a half-second delay between the event generation and the application's query. Also, what happens if the rate between the gateway and the application is 100 Kbps?

Solution: The response delay (D_{resp}) for a 200-byte packet is:

$$D_{resp} = L_{res} \left(\frac{1}{R_{broker}} + \frac{1}{R_{GW}} + \frac{1}{R_{sensor}} \right) = 200 \times 8 \times \left(2 \times 10^{-9} + \frac{10^{-3}}{100} \right) = 16 \text{ ms}$$

The request delay (D_{req}) for a 100-byte packet is:

$$D_{req} = L_{req} \left(\frac{1}{R_{broker}} + \frac{1}{R_{GW}} + \frac{1}{R_{Sensor}} \right) = 100 \times 8 \times \left(2 \times 10^{-9} + \frac{10^{-3}}{100} \right) = 8 \text{ ms}$$

The overall delay is $D = D_{req} + D_{resp} + 500 = 524$ ms. If

$R_{sensor} = R_{GW} = R_{broker} = 100$ Kbps:

$$D_{resp} = L_{res} \frac{3}{R_{broker}} = 200 \times 24 \times \frac{10^{-3}}{100} = 48 \text{ ms}$$

$$D_{req} = L_{req} \frac{3}{R_{broker}} = 100 \times 24 \times \frac{10^{-3}}{100} = 24 \text{ ms}$$

$$D = D_{req} + D_{resp} + 500 = 24 + 48 + 500 = 572 \text{ ms}$$

Exercise: MQTT Transmission Rate

Problem: Consider a sensor that transmits data periodically. Assuming an IEEE 802.15.4 transmission rate of 250 kbps and an average packet size of 200 bytes, how quickly are messages generated using MQTT QoS 0, 1, and 2 sessions?

Solution: For QoS 0, the transmission rate [*message/s*] is:

$$R_{QoS0} = \frac{250 \times 10^3}{8 \times 200} = 156,25$$

In QoS 1, the transmission rate [*message/s*] is:

$$R_{QoS1} = \frac{250 \times 10^3}{2 \times 8 \times 200} = 78,125$$

In QoS 2, the transmission rate [*message/s*] is:

$$R_{QoS2} = \frac{250 \times 10^3}{4 \times 8 \times 200} = 39,062$$

Exercise: CoAP Network Delay

Problem: Consider an end-to-end non-confirmable CoAP scenario where an application requests a readout from a sensor. Given an IEEE 802.15.4 transmission rate of 250Kbps, Ethernet/IPv4 rate of 1 Gbps, $L_{UDP} = 8$ byte, $L_{GET} = 20$ byte, $L_{2.05Content} = 10$ byte, $L_{IPv4} = 20$ byte, $L_{6LoWPAN} = 10$ byte, $L_{eth} = 16$ byte, and $L_{IEEE802.15.4} = 10$ byte, how long does it take for the application to receive the readout?

Solution: The overall delay is the sum of request and response times $T = T_1 + T_2$. The total delay formula is calculated as:

$$T = T_1 + T_2 = \frac{8(16 + 20 + 8 + 20)}{10^9} + \frac{8(10 + 10 + 8 + 20)}{250000} + \frac{8(16 + 20 + 8 + 10)}{10^9} + \frac{8(10 + 10 + 8 + 10)}{250000}$$

$$T = T_1 + T_2 = 1.54 + 1.22 = 2.76 \text{ ms}$$

Exercise: Flooding delay calculation

Problem: Given the following network topology, the transmission/propagation of a single datagram takes 1 s per hop. How long does it take for a datagram to go from device A to device B if the routing is WSN flooding?

Solution: In flooding protocol, the minimum distance from A to B is 3 hops. Therefore, the delay will be 3s.

Exercise: Gossiping / Flooding transmission delay in a ring topology

Problem: In a wireless sensor network with a ring topology where devices send packets, assume that packets are transmitted uniformly to provide full connectivity among all devices. If the transmission rate is 250 Kbps, the packet size is 200 bytes, and collisions and delays are ignored except for transmission delay, what is the expected transmission delay in the flooding scenario?

Solution: The transmission delay of a packet on a single hop is:

$$D_h = \frac{8 \times 200 \times 10^{-3}}{250} = 6.4ms$$

Each device sends to $N = 4$ other devices and therefore the delay for two of these devices is D_h , while the delay for the other two devices is $2D_h$.

The expected delay due to flooding is:

$$D_{flooding} = \frac{2 \times D_h + 2 \times 2 \times D_h}{N} = 9.6ms$$

Exercise: LEACH cluster head selection

Problem: A WSN has five sensors (A, B, C, D, E) with the following battery levels $P_A = 94\%$, $P_B = 97\%$, $P_C = 99\%$, $P_D = 91\%$, and $P_E = 4\%$. In the last 4 rounds, sensors A, B, C, and D have been selected as CH. For the next round, considering LEACH routing, which sensor will likely be selected as CH? Assume that the CH probability is $P = 1/5$.

Solution: Since LEACH does not consider energy levels, the CH will be node E.

June Exam Exercises

Exercise 1: TDMA and FDMA Performance

Question: In a LAN, comprised of $N = 20$ stations with an offered traffic per station, $\lambda = 30$ IU/s, with IU packet length $L = 1000$ byte, and assuming a network capacity of $C = 5$ Mbps, we want to compute the throughput and the time delay characterizing the TDMA and FDMA multiple-access techniques.

Given Parameters:

- N : 20 stations

- λ : 30 IU/s/station
- L : 1000 bytes/IU = 8000 bits/IU
- C : 5 Mbps = 5,000,000 bits/s (bps)

1. Throughput Computation For both TDMA and FDMA, assuming an ideal medium with no packet loss, the network throughput (S or ρ) is normalized to the total network capacity. It is the product of the total arrival rate and packet length, divided by the capacity:

$$S = \rho = \frac{N \cdot \lambda \cdot L}{C}$$

$$S = \frac{20 \text{ stations} \cdot 30 \text{ IU/s/station} \cdot 8000 \text{ bits/IU}}{5,000,000 \text{ bits/s}}$$

$$S = \frac{4,800,000 \text{ bits/s}}{5,000,000 \text{ bits/s}} = \mathbf{0.96}$$

(The normalized throughput is 0.96, or 96% channel utilization. The absolute throughput is 4.8 Mbps).

2. Time Delay Computation First, we compute the base transmission delay (T_t) of a single IU on the full channel:

$$T_t = \frac{L}{C} = \frac{8000 \text{ bits}}{5,000,000 \text{ bits/s}} = 0.0016 \text{ s} = \mathbf{1.6 \text{ ms}}$$

- **TDMA Delay (D_{TDMA}):** The total average delay includes the transmission time, queuing time, and the slot waiting time. The frame duration is $N \cdot T_t = 32 \text{ ms}$.

$$D_{TDMA} = T_t + \left(\frac{\rho}{2(1-\rho)} \cdot (N \cdot T_t) \right) + \frac{1}{2}(N \cdot T_t)$$

$$D_{TDMA} = 1.6 \text{ ms} + \left(\frac{0.96}{2(1-0.96)} \cdot 32 \text{ ms} \right) + \left(\frac{1}{2} \cdot 32 \text{ ms} \right)$$

$$D_{TDMA} = 1.6 \text{ ms} + \left(\frac{0.96}{0.08} \cdot 32 \text{ ms} \right) + 16 \text{ ms}$$

$$D_{TDMA} = 1.6 \text{ ms} + 384 \text{ ms} + 16 \text{ ms} = \mathbf{401.6 \text{ ms}}$$

- **FDMA Delay (D_{FDMA}):** In FDMA, the station's frequency slice is always open (so slot waiting time is 0), but the physical transmission takes N times

longer because the bandwidth is sliced.

$$D_{FDMA} = (N \cdot T_t) + \left(\frac{\rho}{2(1 - \rho)} \cdot (N \cdot T_t) \right)$$

$$D_{FDMA} = 32 \text{ ms} + \left(\frac{0.96}{2(1 - 0.96)} \cdot 32 \text{ ms} \right)$$

$$D_{FDMA} = 32 \text{ ms} + 384 \text{ ms} = \mathbf{416 \text{ ms}}$$

Exercise 1.a: Aloha and Slotted-Aloha Performance

Question: for the same LAN configuration, compute the normalized throughput and the delay in case of Aloha and Slotted-Aloha. Notice that you can consider the approximation $N \rightarrow \infty$. Also, assume that the ratio between the propagation time and the transmission time is $a = 0.01$, and the backoff constant is $K = 4$.

Additional Parameters:

- a (Propagation-to-Transmission ratio): 0.01
- K (Backoff constant): 4
- G (Offered traffic): Equals our throughput S from the previous step (0.96) because $G = N\lambda \frac{L}{C}$.

1. Normalized Throughput

- **Aloha:**

$$S_{Aloha} = G \cdot e^{-2G}$$

$$S_{Aloha} = 0.96 \cdot e^{-2(0.96)} = 0.96 \cdot e^{-1.92} \approx 0.96 \cdot 0.1466 \approx \mathbf{0.1407} \text{ (14.07\%)}$$

- **Slotted-Aloha:**

$$S_{Slotted} = G \cdot e^{-G}$$

$$S_{Slotted} = 0.96 \cdot e^{-0.96} \approx 0.96 \cdot 0.3829 \approx \mathbf{0.3676} \text{ (36.76\%)}$$

2. Delay Computation We first calculate the normalized delay (D_n), then multiply it by the base transmission time T_t (1.6 ms) to find the absolute delay.

- **Aloha Delay:**

$$D_n = (e^{2G} - 1) \cdot \left(\frac{K+1}{2} + 2a \right) + 1 + a$$

$$D_n = (e^{1.92} - 1) \cdot \left(\frac{4+1}{2} + 2(0.01) \right) + 1 + 0.01$$

$$D_n = (5.8209) \cdot (2.52) + 1.01 = 14.668 + 1.01 = 15.678$$

$$D_{Aloha} = 15.678 \cdot 1.6 \text{ ms} = \mathbf{25.08 \text{ ms}}$$

- **Slotted-Aloha Delay:**

$$D_n = (e^G - 1) \cdot \left(\frac{K+1}{2} + 2a \right) + 1.5 + a$$

$$D_n = (e^{0.96} - 1) \cdot (2.52) + 1.51$$

$$D_n = (1.6117) \cdot (2.52) + 1.51 = 4.061 + 1.51 = 5.571$$

$$D_{Slotted} = 5.571 \cdot 1.6 \text{ ms} = \mathbf{8.91 \text{ ms}}$$

Exercise 2: MQTT IoT Architecture

Question: A smart building employs an MQTT-based IoT architecture to monitor environmental conditions. Three sensors are deployed and each publishes measurements every 10 seconds to an MQTT broker. We assume:

- Average payload size = 100 bytes.
- MQTT header size = 10 bytes.
- No packet losses occur.
- ACK and control packet size is 4 bytes.

It is asked to compute:

- The message generation rate (messages/s) of the entire system.
- The total traffic generated by the sensors (bytes/s).
- The total MQTT traffic generated per minute by all sensors.
- Assuming an IEEE 802.15.4 transmission rate of 250 kbps, which and how quickly are messages generated using MQTT QoS 0, 1, and 2 sessions? Ignore propagation delay.

a) Message generation rate (messages/s)

- Rate per sensor = $\frac{1 \text{ message}}{10 \text{ s}} = 0.1 \text{ msg/s}$
- Total system rate = $3 \text{ sensors} \cdot 0.1 \text{ msg/s/sensor} = \mathbf{0.3 \text{ msg/s}}$

b) Total traffic generated by the sensors (bytes/s)

- Total Data Frame size = Payload (100 bytes) + Header (10 bytes) = 110 bytes/msg.
- Total traffic = $0.3 \text{ msg/s} \cdot 110 \text{ bytes/msg} = \mathbf{33 \text{ bytes/s}}$

c) Total MQTT traffic generated per minute by all sensors

- Total traffic per minute = $33 \text{ bytes/s} \cdot 60 \text{ s/minute} = \mathbf{1980 \text{ bytes/minute}}$

d) Message generation and speed using MQTT QoS sessions

First, we find the physical transmission time for the data frames (T_{data}) and the control frames (T_{ctrl}) over the 250 kbps channel:

- $T_{data} = \frac{110 \text{ bytes/msg} \cdot 8 \text{ bits/byte}}{250,000 \text{ bits/s}} = \frac{880 \text{ bits}}{250,000 \text{ bits/s}} = 0.00352 \text{ s} = \mathbf{3.52 \text{ ms}}$
- $T_{ctrl} = \frac{4 \text{ bytes/msg} \cdot 8 \text{ bits/byte}}{250,000 \text{ bits/s}} = \frac{32 \text{ bits}}{250,000 \text{ bits/s}} = 0.000128 \text{ s} = \mathbf{0.128 \text{ ms}}$

Next, we map these times to the specific message sequences required by each QoS level:

QoS Level	Transaction Mechanics	Delay Calculation	Final Delay
QoS 0	PUBLISH only (Fire and Forget)	T_{data}	3.52 ms
QoS 1	PUBLISH + PUBACK (At Least Once)	$T_{data} + T_{ctrl}$	3.648 ms
QoS 2	PUBLISH + PUBREC + PUBREL + PUBCOMP (Exactly Once)	$T_{data} + 3(T_{ctrl})$	3.904 ms